Evaluating COVID-19 Death Trends and Regional Relative Risks using Negative Binomial Regression

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Abstract

This study analyzes COVID-19 mortality factors in the United States (2021–2025) using CDC data. It evaluates demographic, regional, and temporal trends while comparing Poisson and Negative Binomial regression models. Results show that males experienced higher mortality (54.86%), with deaths peaking annually in January. Due to significant overdispersion in the data, the Negative Binomial model provided a superior fit over the Poisson model. Key findings indicate that Region 4 faced the highest mortality risk (IRR = 1.45), while overall death rates significantly declined by 2025. These results underscore the necessity of using appropriate statistical models to address overdispersion in public health surveillance.

Introduction

- This study analyzes the factors associated with COVID-19 mortality in the United States between 2021 and 2025.
- The data is sourced from the Centers for Disease Control and Prevention (CDC) specifically the provisional death counts by month, jurisdiction, and demographic characteristics.

Objective

- To analyze and identify factors associated with COVID-19 mortality in the United States (2021–2025), including demographic, regional and temporal patterns
- To examine temporal patterns and trends in COVID-19 deaths.
- To compare the performance and goodness-of-fit between Poisson regression and Negative Binomial regression models in analyzing overdispersed COVID-19 death count data.

Materials and Methods

- The study utilized provisional COVID-19 death count data from the Centers for Disease Control and Prevention (CDC). The analysis covered a five-year period from January 2021 to 2025 across the United States.
- Dependent Variable: Monthly COVID-19 death counts.
- Independent Variables: Gender (Male/Female), geographic location (10 HHS Regions), and time (Year and Month).
- Time Series Analysis: Applied Time Series Decomposition to separate the mortality data into its Trend, Seasonal, and Residual components to identify cyclical patterns and long-term directions.
- Statistical Modeling: Descriptive Statistics: Used to summarize sample characteristics; group differences were tested using Pearson's chi-square test.
  - Poisson Regression: Initially applied to estimate Incidence Rate Ratios (IRR).
  - Negative Binomial Regression: Implemented as the primary model to address **overdispersion** (where variance exceeds the mean), ensuring more accurate statistical inference.
- Evaluation and Significance: Model goodness-of-fit was evaluated by comparing the Poisson and Negative Binomial results.
- Statistical significance was defined as a p-value < 0.05, with results reported alongside 95% Confidence Intervals (CI).

Conclusion

this study confirms that the Negative Binomial regression model is more appropriate than the Poisson model for analyzing COVID-19 mortality data, as it effectively accounts for significant overdispersion. The findings highlight clear demographic and regional disparities, with males and residents of Region 4 facing the highest mortality risks. Additionally, time series analysis identifies a consistent seasonal peak in January. Despite these periodic surges, an overall declining trend in deaths was observed from 2021 through 2025. These results emphasize the necessity of using robust statistical frameworks to inform targeted public health interventions.

References

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Paul, R., Arif, A. A., Adeyemi, O., Ghosh, S., & Han, D. (2021). Progression of COVID-19 from urban to rural areas in the United States: A spatiotemporal analysis of prevalence rates. The Journal of Rural Health, 37(4), 641–651.

Result and Discussion

Table 1: Characteristic

	Female	Male	Total	*p*
N (%)	384,919 (45.14)	467,725 (54.86)	852,644 (100)	
Region, N (%)				<0.001*
• Region 1	14,080 (3.66)	15,949 (3.41)	30,029 (3.52)	
• Region 2	32,233 (8.37)	36,466 (7.80)	68,699 (8.06)	
• Region 3	39,550 (10.27)	44,740 (9.57)	84,290 (9.89)	
• Region 4	96,180 (24.99)	113,039 (24.17)	209,219 (24.54)	
• Region 5	62,589 (16.26)	73,654 (15.75)	136,243 (15.98)	
• Region 6	52,092 (13.53)	66,713 (14.26)	118,805 (13.26)	
• Region 7	16,438 (4.27)	19,417 (4.15)	35,855 (4.21)	
• Region 8	10,349 (2.69)	13,728 (2.94)	24,077 (2.82)	
• Region 9	49,236 (12.79)	68,262 (14.59)	117,498 (13.78)	
• Region 10	12,172 (3.17)	15,757 (3.36)	27,929 (3.28)	
Month, N (%)				<0.001*
• January	98,178 (25.51)	120,500 (25.76)	218,678 (25.65)	
• February	51,753 (13.45)	65,049 (13.91)	116,802 (13.70)	
• March	24,153 (6.27)	28,806 (6.16)	52,959 (6.21)	
• April	15,727 (4.09)	18,335 (3.92)	34,062 (3.99)	
• May	13,446 (3.49)	15,231 (3.26)	28,677 (3.36)	
• June	10,641 (2.76)	11,935 (2.55)	22,576 (2.65)	
• July	14,424 (3.75)	16,573 (3.54)	30,997 (3.64)	
• August	32,762 (8.51)	40,624 (8.69)	73,386 (8.61)	
• September	39,044 (10.14)	48,243 (10.31)	87,287 (10.24)	
• October	28,391 (7.38)	34,471 (7.37)	62,862 (7.37)	
• November	23,480 (6.10)	28,220 (6.03)	51,700 (6.06)	
• December	32,920 (8.55)	39,738 (8.50)	72,658 (8.52)	

The Pearson Chi-Square test is considered statistically significant when the P-value ⩽ 0.05

Table 2: Poisson Regression Vs Negative Binomial Regression Analysis of Factors Associated with COVID-19 Mortality in the United States, 2021–2025

Characteristics	Poisson Regression		Negative Binomial Regression	
	IRR (95% CI)	P-value	IRR (95% CI)	P-value
Gender, N (%)				
• Male	1.24 (1.23 - 1.24)	<0.001*	1.14 (1.02 - 1.18)	<0.001*
• Female	Ref.	-	Ref.	-
Region, N (%)				
• Region 1	1.03 (1.01 - 1.05)	<0.001*	1.13 (0.87 - 1.47)	0.353
• Region 2	1.22 (1.21 - 1.24)	0.001*	1.17 (0.90 - 1.52)	0.245
• Region 3	1.40 (1.38 - 1.42)	<0.001*	1.31 (1.01 - 1.70)	0.042*
• Region 4	1.60 (1.58 - 1.63)	<0.001*	1.45 (1.12 - 1.89)	0.005*
• Region 5	1.33 (1.31 - 1.36)	<0.001*	1.22 (0.94 - 1.59)	0.128
• Region 6	1.42 (1.40 - 1.44)	<0.001*	1.09 (0.84 - 1.42)	0.507
• Region 7	1.31 (1.29 - 1.33)	<0.001*	1.22 (0.94 - 1.60)	0.134
• Region 8	Ref.	-	Ref.	-
• Region 9	1.19 (1.18 - 1.21)	<0.001*	1.01 (0.78 - 1.31)	0.953
• Region 10	0.99 (0.97 - 1.01)	0.333	1.08 (0.83 - 1.41)	0.562
Month, N (%)				
• January	Ref	-	Ref.	-
• February	0.53 (0.53 - 0.54)	<0.001*	0.58 (0.44 - 0.76)	<0.001*
• March	0.24 (0.24 - 0.24)	<0.001*	0.33 (0.25 - 0.44)	<0.001*
• April	0.16 (0.15 - 0.16)	<0.001*	0.21 (0.15 - 0.27)	<0.001*
• May	0.13 (0.13 - 0.13)	<0.001*	0.16 (0.12 - 0.22)	<0.001*
• June	0.10 (0.10 - 0.10)	<0.001*	0.14 (0.10 - 0.18)	<0.001*
• July	0.14 (0.14 - 0.14)	<0.001*	0.18 (0.13 - 0.24)	<0.001*
• August	0.33 (0.33 - 0.34)	<0.001*	0.30 (0.23 - 0.41)	<0.001*
• September	0.40 (0.39 - 0.40)	<0.001*	0.36 (0.27 - 0.47)	<0.001*
• October	0.29 (0.28 - 0.29)	<0.001*	0.31 (0.23 - 0.41)	<0.001*
• November	0.24 (0.23 - 0.24)	<0.001*	0.26 (0.20 - 0.35)	<0.001*
• December	0.33 (0.33 - 0.33)	<0.001*	0.35 (0.26 - 0.46)	<0.001*

IRR = Incidence Rate Ratio from Poisson regression model; CI = confidence interval.

Reference categories were female, Region 8, year 2021, and January.

Model statistics (Poisson Regression): AIC = 176997.71; BIC = 160093.01, Log likelihood = -88472.85;

Model statistics (Negative Binomial Regression): AIC = 15764.32; BIC = -7917.48; Log likelihood = -7856.16;

Deviance goodness-of-fit p < 0.001; Pearson χ^2/df = 152.8. The Poisson model demonstrated poor goodness-of-fit and substantial **overdispersion** (Pearson χ^2/df >> 1), indicating violation of the equidispersion assumption. *p < 0.05.

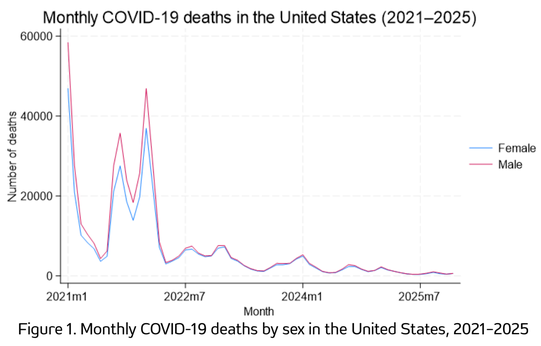

Figure 1. Monthly COVID-19 deaths by sex in the United States, 2021–2025

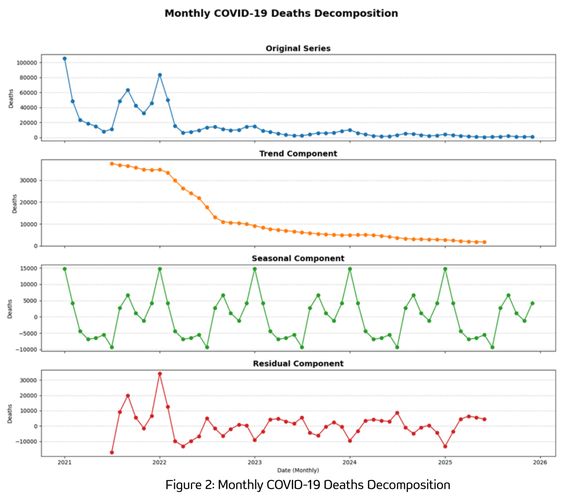

Figure 2: Monthly COVID-19 Deaths Decomposition

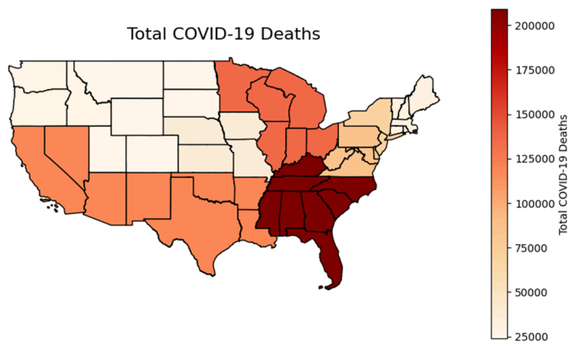

Figure 3. Total COVID-19 Deaths by HHS Region, United States (2021–2025)

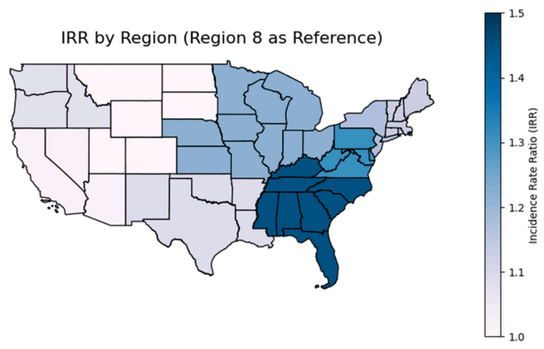

Figure 4. Incidence Rate Ratios (IRR) of COVID-19 Mortality by Region (Region 4 as Reference)